

Student Companion Web Application

A PROPOSAL PRESENTED BY:

AN. AFHAM (SAM/IT/2324/F/0248)

Supervisor:

Mr. M.A. Azrath Areeff

*in partial fulfillment of the requirement
for the award of the*

**HIGHER NATIONAL DIPLOMA OF INFORMATION
TECHNOLOGY**

**ADVANCED TECHNOLOGICAL INSTITUTE SAMMANTHURAI
SLIATE**

2026

TABLE OF CONTENTS

LIST OF FIGURES	iii
TIME FRAME.....	iv
LIST OF ABBREVIATIONS	5
1.INTRODUCTION	1
1.1 Problem in brief.....	2
1.2 Functional Requirement.....	3
1.3 Scope	4
2.PROJECT OVERVIEW	4
2.1 Background.....	5
2.2 Motivation	6
3.AIM AND OBJECTIVE.....	7
3.1 Aim	7
3.2 Objective	7
4.LITERATURE REVIEW.....	8
4.1 STUDY OF SIMILAR APPLICATION	9
4.2 Methodology	10
<i>Figure 1 Methodology</i>	10
4.4 Targeted stakeholders	11
5.PROJECT PLAN	11
5.1 Time plan.....	11
5.3 Gantt Chart	12
6. SYSTEM REQUIREMENTS.....	12
6.1 Software requirements.....	12
6.2 Hardware requirements	12
7.CONCLUSION	13
8. REFERENCES	14

LIST OF FIGURES

Figure 1 Methodology	10
Figure 2 Gantt Chart.....	12

TIME FRAME

Table 1 Time plan	11
--------------------------------	-----------

LIST OF ABBREVIATIONS

ATI	Advance Technological Institute.
SLIATE	Sri Lanka Institute of Advance Technological Education.

1.INTRODUCTION

In today's digital era, students increasingly rely on technology to manage their academic activities and improve productivity. However, many students still depend on scattered tools such as notebooks, messaging apps, and multiple websites to manage tasks, deadlines, study materials, and academic discussions. This lack of a centralized system leads to poor organization, missed deadlines, and inefficient collaboration among students.

The **Student Companion Web Application** is a web-based system designed to centralize and simplify student academic management. The application allows users to manage daily tasks, upload and share study materials, and interact with peers through a Question & Answer system. Additionally, the system includes a Reminder feature, enabling students to set date and time-based alerts for important deadlines, ensuring that no academic responsibility is overlooked.

The system supports role-based access, where juniors can ask questions and manage their tasks, while seniors and administrators can provide answers and guide other students. This creates a collaborative learning environment within the platform.

This system is specifically designed for students who need a simple yet effective tool to manage their academic workflow. It utilizes technologies such as HTML and CSS for the frontend, PHP for backend processing, MySQL for database management, and JavaScript for implementing reminder alerts and interactive features.

The proposed system aims to improve student productivity, enhance collaboration between juniors and seniors, provide a centralized platform for academic resources, and ensure better time management through its integrated reminder system.

1.1 Problem in brief

Currently, many students rely on multiple disconnected tools such as notebooks, messaging applications, and different online platforms to manage their academic activities. This traditional and unstructured approach creates several significant challenges

Time Management Issues: Students often struggle to organize and track their daily academic tasks, leading to poor productivity and missed responsibilities.

Missed Deadlines: Without a proper reminder system, students frequently forget assignment deadlines, project submissions, and important academic schedules.

Lack of Centralized Platform: Study materials such as notes, PDFs, and useful links are scattered across different platforms, making it difficult to access them when needed.

Limited Peer Support: Students face delays in getting help for academic doubts due to the absence of a dedicated platform for asking and answering questions.

Poor Collaboration: There is no structured system that connects juniors and seniors for knowledge sharing and guidance.

Difficulty in Resource Management: Managing and retrieving previously shared notes or study materials becomes time-consuming and inefficient.

No Real-Time Assistance: Students cannot quickly access answers or guidance when facing academic difficulties.

Lack of Organization: Academic activities such as tasks, notes, and discussions are not maintained in an organized manner, leading to confusion and inefficiency.

The absence of an integrated academic management system creates inefficiencies that affect students' productivity and learning experience. Therefore, there is a strong need for a modern, centralized, and efficient web-based solution to manage student activities effectively.

1.2 Functional Requirement

The **Student Companion Web Application** will include the following functional requirements:

User Authentication:

- The system shall allow users to log in using email and password.
- The system shall authenticate users and maintain session management.
- The system shall support role-based access (Admin, Senior, Junior).

Task Management:

- The system shall allow users to add new tasks.
- The system shall display all tasks created by the user.
- The system shall allow users to manage and track their daily activities.

Reminder System:

- The system shall allow users to set reminders with date and time.
- The system shall store reminder details in the database.
- The system shall trigger alerts when the reminder time is reached.
- The system shall display all upcoming reminders to the user.

Notes Management:

- The system shall allow users to upload study materials (PDF, documents).
- The system shall allow users to share external links (YouTube, Google Drive, etc.).
- The system shall store uploaded files in the system.
- The system shall allow users to download or access shared resources.

Q&A System:

- The system shall allow junior users to post questions.
- The system shall display all posted questions.
- The system shall allow senior and admin users to provide answers.
- The system shall display answers along with the respective questions.

Role-Based Access Control:

- The system shall restrict access based on user roles.
- Junior users shall not have permission to answer questions.

Dashboard:

- The system shall display user information such as name and role.
- The system shall provide navigation to all modules (Tasks, Notes, Q&A, Reminders).
- The system shall present a simple and user-friendly interface.

Session Management:

- The system shall maintain user sessions after login.
- The system shall restrict access to unauthorized users.

1.3 Scope

This system focuses on improving student academic management by:

Improved Academic Organization: The system will help students manage all academic activities such as assignments, timetables, study materials, and notices in a single platform, reducing confusion and improving productivity.

Enhanced Time Management: Students will be able to track deadlines and schedules effectively, ensuring that no assignments or important academic tasks are missed.

Better Communication: The application will improve communication between students and lecturers by providing a centralized space for updates, announcements, and academic discussions..

Centralized Data Management: All academic information such as notices, study materials, and assignments will be stored in a structured system, making it easy to access anytime from anywhere.

Improved Academic Performance: With proper tracking of tasks and deadlines, students can stay organized and focused, which will contribute to better academic results.

Cost-Effective Solution: The system will be developed using free and open-source technologies such as HTML, CSS, JavaScript, PHP, and MySQL, reducing development and maintenance costs.

User-Friendly Interface: The web application will be designed with a simple and clean interface, making it easy for students and lecturers to use without requiring technical training.

Cross-Platform Accessibility: The system will be accessible through any modern web browser on laptops, desktops, and mobile devices.

Secure Data Handling: User data, academic records, and communications will be securely stored in the database with proper authentication and access control.

2.PROJECT OVERVIEW

2.1 Background

Managing academic activities is a fundamental part of every student's daily life. It includes tracking tasks, remembering deadlines, organizing study materials, and seeking help for difficult subjects. These activities directly impact student productivity, performance, and overall learning experience.

Traditionally, students rely on notebooks, mobile notes, messaging applications, and multiple online platforms to manage these responsibilities. While these methods are easy to use, they suffer from significant limitations in terms of organization, efficiency, and accessibility..

In recent years, web technologies have evolved rapidly, making it easier to develop centralized systems that can handle multiple functionalities within a single platform. Web-based applications provide accessibility from any device with a browser, making them highly convenient for students.

At the same time, the importance of time management has increased in academic environments. Missing deadlines and poor task organization are common problems faced by students. Incorporating a **Reminder System with date and time alerts** can significantly improve time management and reduce missed academic responsibilities.

The **Student Companion Web Application** leverages these modern web technologies to create a centralized academic management platform. The system is developed using **HTML and CSS for the frontend, PHP for backend processing, and MySQL for database management**, with **JavaScript used for interactive features such as reminder alerts**.

This application integrates multiple features such as task management, note sharing, and a Question & Answer system into a single platform. It also introduces role-based interaction, allowing seniors to guide juniors, thereby promoting collaborative learning.

This project demonstrates how modern web development technologies can be used to solve real-world academic challenges. It also serves as a practical implementation of full-stack development concepts, providing valuable experience in building scalable and user-friendly web applications.

2.2 Motivation

The motivation for developing this system stems from several factors:

Personal Experience: As a student, I have observed the difficulties faced in managing academic tasks, remembering deadlines, and organizing study materials. This experience inspired the idea of creating a centralized system to simplify these activities.

Lack of Centralized System: Many students rely on multiple disconnected tools such as notebooks, messaging apps, and different websites. There is a clear need for a single platform that integrates task management, notes, and academic communication.

Need for Better Time Management: Students often miss deadlines due to poor planning and lack of reminders. Introducing a date and time-based reminder system helps improve time management and productivity.

Availability of Web Technologies: Modern web technologies such as PHP, MySQL, HTML, CSS, and JavaScript make it possible to develop a fully functional and interactive system without high costs.

3. AIM AND OBJECTIVE

3.1 Aim

The primary aim of this project is to develop a **web-based Student Companion Application** that helps students manage their academic activities efficiently. The system is designed to centralize task management, note sharing, and academic interaction, while incorporating a **date and time-based reminder system** to reduce missed deadlines and improve time management.

The application aims to eliminate the need for multiple disconnected tools by providing a single, organized platform where students can track tasks, access study materials, and collaborate with peers in real time.

3.2 Objective

The specific objectives of this project are:

Simplify Academic Management:

Provide an intuitive web application that allows students to manage their daily tasks, study materials, and academic activities in one place.

Improve Time Management:

Implement a **date and time-based reminder system** that helps students track deadlines and avoid missing important academic tasks.

Provide User-Friendly Interface:

Design a simple, modern, and easy-to-use interface that requires minimal learning effort for user

Ensure Data Security:

Implement secure login and session management to protect user data and restrict unauthorized access.

Provide Dashboard Overview:

Display essential user information and provide easy navigation to all modules such as Tasks, Notes, Q&A, and Reminders.

4.LITERATURE REVIEW

Today, many students use technology to help with their studies. There are different types of applications such as task managers, note-taking apps, and messaging platforms. These tools help students manage their work, but they are not connected in one place. Because of this, students often feel confused and disorganized.

Some systems allow students to set reminders for their tasks and deadlines. These systems are useful for time management, but students usually forget to use them regularly since they are separate from their study activities.

There are also platforms where students can share notes and study materials. However, these materials are often stored in different places like emails, chats, or cloud storage. This makes it difficult to find important information when needed.

In addition, some websites and applications provide question-and-answer features where students can ask doubts. These systems help students learn from each other. But most of these platforms do not include other important features like task management or reminders.

Many learning management systems (LMS) are used in schools and universities. These systems help teachers share materials and announcements. However, they are sometimes complex and do not focus much on student-to-student interaction, especially between juniors and seniors.

From studying these existing systems, it is clear that there is no single platform that combines all these features together. Students still need to use multiple applications to manage their academic work.

Therefore, the proposed **Student Companion Web Application** is designed to solve these problems. It combines task management, reminders, note sharing, and a question-and-answer system into one simple platform. This will help students stay organized, manage their time better, and easily collaborate with others.

4.1 STUDY OF SIMILAR APPLICATION

1. **Notion – All-in-One Student Workspace-** Notion is a web-based productivity application that helps students manage notes, tasks, and study materials in one place. It provides features such as task tracking, note organization, databases, and real-time collaboration. Students can create personalized dashboards to organize their academic work. However, the system can be complex for beginners and requires manual setup to fully utilize its features. [1]
2. **Todoist – Task Management Application-** Todoist is a task management tool designed to help users organize their daily activities and deadlines. Students can create tasks, set due dates, and receive reminders to manage their academic responsibilities. The system is simple and user-friendly but mainly focuses on task management and does not include features like note sharing or student collaboration. [2]
3. **Google Workspace – Collaboration Platform-**Google Workspace is a cloud-based platform that provides tools such as Google Docs, Google Drive, and Google Calendar for document creation, storage, and collaboration. It allows multiple users to work together in real time and access files from anywhere. However, it does not provide a fully integrated system for managing tasks, reminders, and student interactions in one place. [3]
4. **ClickUp – Productivity and Project Management Tool-** ClickUp is an all-in-one productivity platform that offers features such as task management, reminders, document sharing, and team collaboration. It helps users track their work and improve productivity. However, the system is mainly designed for project management and business use, making it slightly complex for students who need a simple academic tool. [4]
5. **Google Drive – File Storage and Sharing System -** Google Drive is a cloud storage service that allows users to upload, store, and share files such as documents, PDFs, and videos. Students can easily access their study materials from any device. However, it does not include features like task management, reminders, or question-and-answer systems for academic support. [5]

4.2 Methodology

Software Development Life Cycle is the application of standard business practices to building software applications. It's typically divided into six to eight steps. Some project managers will combine, split, or omit steps, depending on the project's scope. These are the core components recommended for all software development projects.

1. Planning
2. Analysis
3. Design
4. Testing
5. Implementation
6. Maintenance

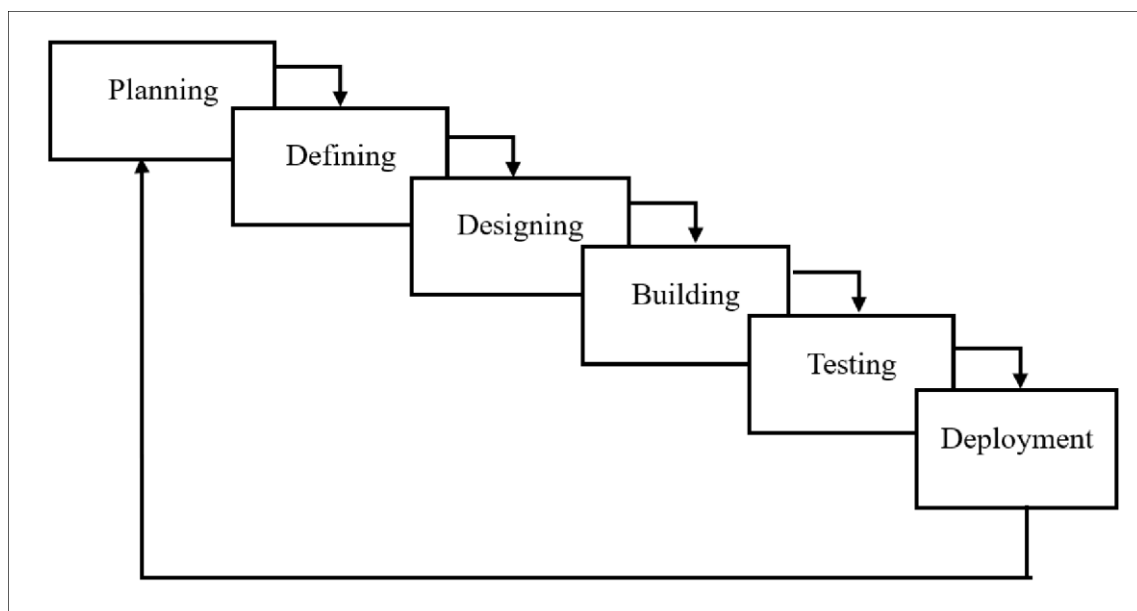


Figure 1 Methodology

4.4 Targeted stakeholders

The targeted stakeholders for this system are

1. Lecturer (Admin)
2. HNDIT Students (Seniors & Juniors)

5.PROJECT PLAN

5.1 Time plan

Item	Date
Project Proposal	20/04/2026
Software Project Management Plan	21/04/2026 to 27/04/2026
UI/UX Design (Wireframes)	28/04/2026 to 04/05/2026
Database Design (Firestore Schema)	05/05/2026 to 11/05/2026
Mid Progress Review	05/05/2026 to 11/05/2026
Development - Authentication & Role-based Setup	12/05/2026 to 18/05/2026
Development - Task Management & Reminder System	19/05/2026 to 25/05/2026
Development - Notes Management (File Uploads)	26/05/2026 to 01/06/2026
Development - Junior-Senior Q&A System	02/06/2026 to 08/06/2026
Development - Dashboard & Reporting	09/06/2026 to 15/06/2026
Software Testing	16/06/2026 to 22/06/2026
Final Report	16/06/2026 to 22/06/2026
Final Project Presentation	22/06/2026

Table 1: Time plan

5.3 Gantt Chart

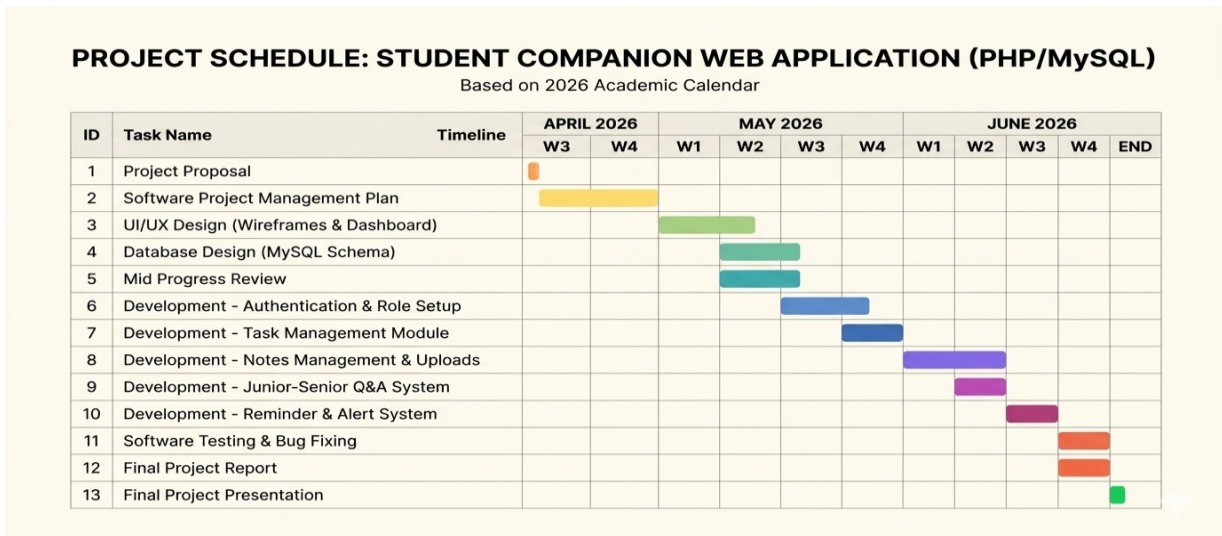


Figure 3 Gantt Chart

6. SYSTEM REQUIREMENTS

6.1 Software requirements

Operating System: Windows 10

Web Server Environment: XAMPP (Apache Server, PHP, MySQL)

Programming Language (Backend): PHP 8+

Frontend Technologies: HTML, CSS, JavaScript

Database: MySQL

Browser: Google Chrome / Microsoft Edge

6.2 Hardware requirements

Processor: Intel Core i5 or equivalent

Memory: 4 GB (8 GB recommended)

Storage: 10 GB free space

7.CONCLUSION

The **Student Companion Web Application** addresses a critical need in the academic life of students: a centralized, efficient, and organized platform to manage their daily educational activities. By integrating task management, study resource sharing, and a peer-to-peer Q&A system, this application provides a one-stop solution that replaces the traditional, scattered methods of tracking academic progress.

The key benefits of this system include:

- **Centralized Organization:** Students no longer need to rely on multiple apps or notebooks; all tasks, notes, and deadlines are managed in a single, secure platform.
- **Improved Time Management:** The integrated date and time-based **Reminder System** ensures that students stay ahead of their schedules and never miss important deadlines.
- **Enhanced Peer Collaboration:** The role-based **Q&A System** fosters a learning community where juniors can seek guidance and seniors can share their knowledge effectively.
- **Easy Access to Resources:** The **Notes Management** module provides a structured way to store and retrieve study materials (PDFs and links), saving time and effort for students.
- **Cost-Effective and Accessible:** Built using open-source technologies like PHP, MySQL, and JavaScript, the system is lightweight, easy to maintain, and accessible through any web browser.
- **User-Friendly Experience:** With a clean and intuitive dashboard, users can navigate between different modules without any technical training.

This project successfully demonstrates how full-stack web development can be applied to solve real-world academic challenges. It not only improves the productivity of individual students but also strengthens the academic bond between the student community through technology.

Future Enhancements:

In the future, this system could be expanded to include:

Real-time Chat: A direct messaging system for instant communication between users.

Mobile App Version: A dedicated mobile application for better accessibility and push notifications.

8. REFERENCES

- [1] " 01 "Notion – The All-in-One Workspace for Your Notes, Tasks, and Collaboration "I. Zhao .Available: <https://www.notion.so> .[متصل]. MARCH 2016
- [2] A. S. (. Inc.), "odoist – Organize Your Work and Life," 01 January 2007. [Online]. Available: <https://todoist.coM>.
- [3] G. LLC, "Google Workspace – Business Apps and Collaboration Tools," 28 August 2006. [Online]. Available: <https://workspace.google.com>.
- [4] Z. E. & A. Yurkowski, "ClickUp – One App to Replace Them All," 1 october 2017. [Online]. Available: <https://clickup.com>.
- [5] G. LLC, "Google Drive – Cloud Storage & File Sharing Platform," 24 April 2012. [Online]. Available: <https://drive.google.com>.